MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON
TYPES 2N2034, 2N2858, 2N2859, 2N2911, and
TX2N2034, TX2N2858, TX2N2859, TX2N2911

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for silicon, NPN, transistors for particular use in amplifier and switching electronic-circuit applications. (See 3.4 and 6.2 herein.) The prefix "TX" identifies devices meeting the special process-conditioning, testing, and screening requirements in 4.4 herein, and the stringent quality-control (LTPD) requirements in Tables I, II, and III herein.

1.2 Outline and dimensions.- See Fig. 1 herein. (TO-5)

1.3 Maximum ratings.- (At T_A=+25°C, unless otherwise specified):

	P_T 1/ at: T_A=+25°C	P_T 2/ at: T_C=+100°C	V_{CBO}	V_{CEO}	V_{CEX}	V_{EBO}	I_C	T_{stg} and T_J(oper)
	W	W	Vdc	Vdc	Vdc	Vdc	Adc	°C
2N2034	1	7.5	80	60	80	10	3	-65 to +200
2N2858	1	7.5	100	80	100	10	3	-65 to +200
2N2859	1	7.5	120	100	120	10	3	-65 to +200
2N2911	1	7.5	150	125	150	10	3	-65 to +200

1/ For T_A>+25°C, derate linearly at 5.71 mW/°C.

2/ For T_C>+100°C, derate linearly at 75 mW/°C.

1.4 Particular electrical characteristics.- (At T_A=+25°C, unless otherwise specified):

1/ h_{FE} at: V_{CE}=4 Vdc, I_C=1 Adc		$\lvert h_{fe} \rvert$ at: f=1 MHz, V_{CE}=4 Vdc, I_C=0.1 Adc		1/ $V_{BE(sat)}$ at: I_C=1 Adc, I_B=0.1 Adc		1/ $V_{CE(sat)}$ at: I_C=1 Adc, I_B=0.1 Adc		Switching			
								t_{on}		t_{off}	
								(See Table I and Fig. 3 herein.)			
All Types		All Types		All Types		All Types		All Types			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
---	---	---	---	Vdc	Vdc	Vdc	Vdc	usec	usec	usec	usec
20	60	1	6	---	1.2	---	0.3	---	3	---	6

1/ Pulsed (see 4.3.1 herein).

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
MILITARY
MIL-S-19500 Semiconductor Devices, General Specification For

STANDARDS
MILITARY
MIL-STD-202 Test Methods For Electronic And Electrical Component Parts
MIL-STD-750 Test Methods For Semiconductor Devices
MIL-STD-1276 Leads, Weldable, For Electronic Component Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements.- Requirements for the transistors shall be in accordance with Specification MIL-S-19500, and as otherwise specified herein.

3.2 Abbreviations and Symbols.- The abbreviations and symbols used herein are defined in Specification MIL-S-19500, and as follows:

$$t_{off} \ldots\ldots (t_s + t_f)$$

3.3 Design and construction.- The transistors shall be of the design, construction, and physical dimensions specified in Figure 1.

3.3.1 Terminal arrangement.- The terminal arrangement on the transistors shall be as indicated in Figure 1.

3.3.2. Operating position.- The transistor shall be capable of proper operation in any position.

3.3.3 Terminal-lead length.- Terminal-lead length(s) other than that established in Figure 1 may be furnished when specified in the contract or order (see 6.3 herein) where the devices covered herein are required directly for particular equipment-circuit installation. Where such other lead-lengths are required and provided, it shall not be construed as affecting adversely the Qualified-product status of the device, or applicable JAN marking.

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3.3.4 Terminal-lead material and finish.- The terminal-lead material shall be Type K (Kovar) or Type F (Alloy 52) conforming to requirements in Standard MIL-STD-1276. The terminal-lead finish shall be gold (per pertinent requirements in Standard MIL-STD-1276); however, if so specified in the contract or order, the lead-material finish may be tin, or, tin-coating over gold. This tin-finish requirement shall not be construed as affecting adversely the qualified-product status of the device, or applicable JAN marking. (See 6.3b herein.)

3.3.4.1 Selectivity of terminal-lead material.- Where choice of lead material (see 3.3.4 above) is desired, stipulation for the particular material shall be made in contract or order. (See 6.3c herein.)

3.4 Performance characteristics.- The transistor performance characteristics shall be as specified in Tables I, II, and III herein. Except where specifically differentiated for respective transistor types (see 1.3, 1.4, and Tables I, II, and III herein), the performance requirements, including operating characteristics, ratings, test conditions, and test limits, apply equally to all transistor types covered herein.

3.4.1 "TX" special requirements.- Specified "TX" device-types shall, in addition to all other performance requirements herein, be process-conditioned, tested, and screened in accordance with 4.4 herein. (At option of the manufacturer, the non-TX device-types may be subjected to the same process-conditioning and testing. See 3.5.1 herein.)

3.5 Marking.- Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the "classification" type designation only. The "Manufacturer's identification" and "country of origin" may, at option of the manufacturer, be omitted from being marked directly on the semiconductor device covered herein.

3.5.1 "TX" identification marking.- Marking of devices conforming to "TX" and all other applicable requirements herein (see 3.4.1) shall include the prefix "TX" immediately preceding the type designation.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and acceptance inspection.- Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Groups A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II and III, respectively, herein. Quality Conformance inspection shall include inspection of Preparation for Delivery (see 5.1 herein).

4.2.1 Specified LTPD for subgroups.- The LTPD specified for a subgroup in Tables I, II, and III herein shall apply for all of the tests, combined, in the subgroup.

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4.2.2 Qualification testing.- The non-TX device types shall be considered subject to Qualification tests. It shall be understood that procurement of the TX device types will be made from sources for the approved non-TX devices as listed in Qualified Products List (QPL)-19500. Upon request and validation by the qualifying activity, qualification may be extended (and appropriate QPL-listing effected) to TX devices manufactured by approved sources for the non-TX devices.

4.2.2.1 Qualification relative to lead-material-finish.- Devices constructed with either of the lead-material finishes established herein (see 3.3.4) may be submitted for Qualification approval. The type of finish used by the manufacturer shall be stipulated in the submittal data for the product to be tested.

4.2.3 Special Group A criteria for Qualification inspection.- For Qualification inspection, only 10 failures will be permitted for all Group A tests combined. Hereto, where the manufacturer deems that the data in paragraph 4.4.2 of Specification MIL-S 19500 are invokable, notification shall be made to the Chief, Components and Materials Standardization Branch, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703, attention: AMSEL-PP-EM-2.

4.2.4 Group B-Group C life test samples.- Samples of non-TX transistor types, only, that have been subjected to Group B, 340-hour life test may be continued on test for 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subjected to the Group C, 1000-hour evaluation after they have passed the Group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1000 hours shall be computed for 1000-hour acceptance. (See 4.2.5 below.) "TX" transistor types shall be subjected to 1000-hour life tests under Group B inspection requirements, and the lambda (λ) established thereto in Table II shall apply.

4.2.5 Group C testing.- Group C tests shall be performed on a lot every 6 months. (See Table III herein.) The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of Group C, 6-month, test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required Group C tests.

4.2.6 Disposition of sample units.- Sample units that have been subjected to Group B, Subgroup 2, 4, and 5 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroup 1, 3, 6, 7, 8, 9, and 10, and all Group C tests (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.2.7 Copies of lot-inspection data.- When specified in the contract or order, one copy of the Quality Conformance inspection data, including manufacturer's test data and findings, covering the device inspection lot being shipped shall accompany the device(s) concerned.

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4.3 Particular examination and test requirements.-

4.3.1 "Pulsed"-conditions measurements.- Measurements required herein to be effected under pulsed conditions, shall be made in accordance with "Pulse Measurements" requirements in Section 4 of Standard MIL-STD-750.

4.3.2 Interval for End-Point Test measurements.- All applicable End-Point Test measurements shall be performed within 96 hours after sample units have been subjected to required physical-mechanical or environmental test(s).

4.3.3 Mechanical damage resulting from tests.- Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Groups A, B, or C tests.

4.3.4 Safe Operating Area (DC Operation) test.- Each transistor under test shall sustain the applicable test conditions, and the following acceptance criteria shall apply:

a. I_C(for each transistor) shall not very more than 10% during the dc operation; and

b. All other specified End-Point Test(s) limits shall not be exceeded, after the dc-operation test.

c. Correlation note: Satisfactory endurance of the transistor throughout Test #1, 2, 3 respectively (per Table II, Subgroup 6 herein) is directly associable with ascertainment of the "safe operating area" for the transistors as illustrated in the nomograph of Fig. 5 herein.

4.4 Process conditioning, testing, and screening for "TX" devices.- All "TX" devices (100% of each lot) shall be subjected to the process-conditioning, testing, and screening specified in 4.4.2 through 4.4.7.4 herein, in accordance with the procedural sequence established in Fig. 2 herein, prior to submission of the lot to Group A, B, and C tests. The manufacturer shall permit the authorized Government representative to witness, concurrently with time of manufacturer's performance of these tests, the process-conditioning, testing, and screening on the devices. Those conditioning and screening tests normally performed by a manufacturer as standard-production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than those specified herein.

4.4.1 Quality Assurance records and determinants.- The manufacturer shall maintain and make available for perusal and review at any time, during a period of 3 years minimum, to the authorized Government representative or other contracting customer concerned, complete lot-by lot records for these devices. All devices that fail to meet the process-conditioning, testing, and screening requirements specified herein shall be removed from the respective lot, and the test-failure identity and quantity removed shall be recorded in the permanent lot history. (See 6.3, 6.4, and 6.5 herein.)

4.4.2 High-temperature storage.- The transistors shall be stored for at least 24 hours at a minimum temperature (T_A) of +200°C.

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4.4.3 Thermal shock (temperature cycling).- The transistors shall be subjected to temperature cycling (thermal shock) in accordance with MIL-STD-750, Method 1051, Test Condition C, except that cycling duration shall be 10 continuous cycles and exposure time at temperature extremes shall be 15 minutes minimum.

4.4.4 Reverse-bias burn-in.- The transistors shall be operated, for 12 hours minimum, at T_A=+175°C, under the following test conditions:

2N2034 : V_{CB}=60 Vdc
2N2858 : V_{CB}=80 Vdc
2N2859 : V_{CB}=100 Vdc
2N2911 : V_{CB}=100 Vdc

At end of the 12-hour period, the above-specified test voltages shall be maintained on the transistors until T_A=+30° ± 5°C is reached.

4.4.5 Acceleration.- The transistors shall be subjected to the constant acceleration (centrifugal) test per Method 2006 in Standard MIL-STD-750, except that the following shall apply:

a. Centrifugal force = 20,000 G

b. Orientation = Y1, only.

c. Hold time of 1 minute shall not apply hereto.

4.4.6 Seal (fine-leak) test.- The transistors shall be subjected to the Seal test per Standard MIL-STD-202, Method 112, Test Condition C, procedure III a or III b (see 4.4.6.1 or 4.4.6.2 below), except that testing for gross leaks shall be in accordance with 4.4.6.3 below.

4.4.6.1 Conditions for procedure III a.- The sealed chamber in which the transistors are placed shall be pressurized with helium gas to 50 psig, minimum, for a minimum of 4 hours. At end of that period, the transistors shall be removed from the chamber and, within 30 minutes, be subjected to a helium leak-detection test. Those transistors showing a leakage rate of greater than 1 X 10^{-7} cc/sec of helium, when measured at a differential pressure of 1 atmosphere, shall be rejected and removed from the lot.

4.4.6.2 Conditions for procedure III b.- The devices shall be placed in an activation tank, pressurized with Krypton 85 tracer gas in a nitrogen solution, for sufficient time to detect a leak rate of 1 X 10^{-8} atmospheric cubic centimeters per second (atm cc/sec). Within four hours after subjection to this pressurization, the leak rate of the devices shall be determined on an attribute basis using the general equation shown below. Any device exhibiting a leak rate equal to or greater than 1 X 10^{-8} atm cc/sec shall be removed from the lot. The general equation for use with radioactive-gas leak test equipment is:

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$$Q = \frac{R}{SKT(P_0^2 - P_1^2)}$$

where:

Q = leak rate in atm cc/sec.
R = net counting rate of tested part above background in cts/min.
S = specific activity of the test gas mixture in uCi/atm cc.
K = counting efficiency of the system for the given part in cts/min Ci.
P_0 = pressure of test gas in activation tank during pressurization in atm abs.
P_1 = pressure inside part under test in atm abs.
T = duration of pressurization in test gas mixture in seconds.

4.4.6.3 Gross Leak test.- The transistors shall be immersed for a minimum period of 15 seconds in a suitable, non-corrosive liquid maintained at approximately +100°C, and shall be observed for emanating bubbles. Those transistors from which a bubble trace is observed shall be rejected and removed from the lot.

4.4.7 Burn-in conditioning, tests, and evaluation (screening).- To enable proper evaluation of the transistors conditioned and tested in accordance with the following requirements, each transistor and lot shall be handled or identified in such manner that results for each individual transistor and lot can be ascertained and recorded.

4.4.7.1 Pre burn-in measurements.- The transistors shall be tested for the characteristics listed in Table IV below, and the values found shall be recorded. Those transistors that fail to meet the specified limits shall be rejected and removed from the lot.

Table IV. Pre burn-in measurements.

Test Method per MIL-STD-750	Examination or Test 1/	Test Conditions -	Symbol	Limits Min	Limits Max	Unit
3041	Collector-emitter cutoff current:	Bias Cond.A V_{BE}=-1.5 Vdc				
	2N2034	V_{CE}=80 Vdc	I_{CEX}	---	20	uAdc
	2N2858	V_{CE}=100 Vdc	I_{CEX}	---	20	uAdc
	2N2859	V_{CE}=120 Vdc	I_{CEX}	---	20	uAdc
	2N2911	V_{CE}=150 Vdc	I_{CEX}	---	20	uAdc
3076	Forward-current transfer ratio	V_{CE}=4 Vdc I_C=1 Adc Pulsed; 2/	h_{FE}	20	60	---
3061	Emitter-base cutoff current	Bias Cond.D V_{EB}=10 Vdc	I_{EBO}	---	10	uAdc

1/ See 3.4 herein.
2/ See 4.3.1 herein.

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4.4.7.2 Burn-in conditioning.- The transistors shall be operated under the following conditions for 168 hours, minimum:

T_A=+25°C
P_T=1 W (no heat sink)
2N2034 : V_{CB}= 40 Vdc
2N2858 : V_{CB}= 60 Vdc
2N2859 : V_{CB}= 100 Vdc
2N2911 : V_{CB}= 100 Vdc

4.4.7.3 Post burn-in tests and measurements.- After burn-in conditioning, the transistors shall be permitted to return to room-ambient temperature thermal-equilibrium and retested, within a maximum total elapsed time of 96 hours, for the characteristics listed in Table IV above. The values now found for each individual transistor shall not exceed the following (respectively applicable to the pre burn-in measurements per Table IV above):

ΔI_{CEX}= ±100% or 2uA, whichever is greater.
Δh_{FE}= $^{+25}_{-10}$%
ΔI_{EBO}= ±100% or 1 uA, whichever is greater.

4.4.7.4 Burn-in conditioning evaluation (screening).- Those transistors whose measured characteristics exceed the delta (Δ) limits specified in 4.4.7.3, above, or the limits of Table IV, shall be rejected and removed from the lot. Where 10% or more of the transistors subjected to the Burn-in test are found to exceed those limits, the entire lot shall be considered non-acceptable as "TX" devices.

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Table I. Group A inspection.

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		10	5				
2071	Visual and mechanical examination	---			---	---	---	---
	Subgroup 2		5	3				
3001	Collector-base breakdown voltage:	Bias Cond.D I_C=150 uAdc			---	---	---	---
	2N2034				BV_{CBO}	80	---	Vdc
	2N2858				BV_{CBO}	100	---	Vdc
	2N2859				BV_{CBO}	120	---	Vdc
	2N2911				BV_{CBO}	150	---	Vdc
3011	Collector-emitter breakdown voltage:	Bias Cond.D I_C=100 mAdc Pulsed; 2/						
	2N2034				BV_{CEO}	60	---	Vdc
	2N2858				BV_{CEO}	80	---	Vdc
	2N2859				BV_{CEO}	100	---	Vdc
	2N2911				BV_{CEO}	125	---	Vdc
3041	Collector-emitter cut-off current:	Bias Cond.A. V_{BE}=-1.5 Vdc						
	2N2034	V_{CE}= 80 Vdc			I_{CEX}	---	20	uAdc
	2N2858	V_{CE}=100 Vdc			I_{CEX}	---	20	uAdc
	2N2859	V_{CE}=120 Vdc			I_{CEX}	---	20	uAdc
	2N2911	V_{CE}=150 Vdc			I_{CEX}	---	20	uAdc
3061	Emitter-base cut-off current:	Bias Cond.D V_{EB}=10 Vdc			I_{EBO}	---	10	uAdc

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Table I. Group A inspection.- (Cont'd).

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 3		5	3				
3066	Base-emitter saturation voltage	Test Cond.A I_C=1 Adc I_B=100 mAdc Pulsed; 2/			V_{BE}(sat	---	1.2	Vdc
3071	Collector-emitter saturation voltage	I_C=1 Adc I_B=100 mAdc Pulsed; 2/			V_{CE}(sat	---	0.3	Vdc
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=1.0 Adc V_{CE}=4 Vdc			h_{FE}	20	60	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=0.250 Adc V_{CE}=4 Vdc			h_{FE}	35	200	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=2.0 Adc V_{CE}=4 Vdc			h_{FE}	10	---	---
	Subgroup 4		10	5				
3306	Magnitude of common-emitter small signal short-circuit transfer ratio	I_C=100 mAdc V_{CE}=4 Vdc f=1 MHz			$\|h_{fe}\|$	1	6	---
3236	Output capacitance (open circuit)	V_{CB}=15 Vdc I_E=0 f=0.1 to 1.0 MHz			C_{obo}	---	200	pf
3251	Pulse response:	Test Cond.A except test circuits and pulse req'ts per Fig. 3 herein						
	Turn-on time				t_{on}	---	3	usec
	Turn-off time				t_{off}	---	6	usec

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Table I. Group A inspection (Cont'd).

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 5		10	5				
3/	High-temperature operation:	T_A =+150°C 4/						
3041	Collector-emitter cutoff current:	Bias Cond.A V_{BE}=-1.5 Vdc						
	2N2034	V_{CE}= 80 Vdc			I_{CEX}	---	150	uAdc
	2N2858	V_{CE}=100 Vdc			I_{CEX}	---	150	uAdc
	2N2859	V_{CE}=120 Vdc			I_{CEX}	---	150	uAdc
	2N2911	V_{CE}=150 Vdc			I_{CEX}	---	250	uAdc
---	Low-temperature operation:	T_A=-55°C 4/						
3076	Forward-current transfer ratio	Pulsed; 2/ I_C=1 Adc V_{CE}=4 Vdc			h_{FE}	20	110	---

1/ See 3.4 herein. 2/ See 4.3.1 herein.

3/ In this Subgroup, the sample units subjected to the High-Temperature Operation test shall be permitted to return to and be stabilized at room ambient temperature prior to their being subjected to the Low-Temperature Operation test.

4/ Measurement(s) shall be made after thermal equilibrium has been reached at the temperature specified.

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Table II. Group B inspection.

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		20	20				
2066	Physical dimensions	---			---	---	---	---
	Subgroup 2		15	15				
2026	Solderability	---			---	---	---	---
1051	Temperature cycling	Test Cond.C			---	---	---	---
1056	Thermal shock (glass strain)	Test Cond.A			---	---	---	---
2/	Seal (leak rate)	Test Cond.C, procedure III; Test Cond.A for gross leaks			---	---	10_q^{-7}	atm/cc/sec
1021	Moisture resistance	---			---	---	---	---
	End-Point tests:							
3041	Collector-emitter cutoff current:	Bias Cond.A V_{BE}=-1.5 Vdc						
	2N2034	V_{CE}= 80 Vdc			I_{CEX}	---	20	uAdc
	2N2858	V_{CE}=100 Vdc			I_{CEX}	---	20	uAdc
	2N2859	V_{CE}=120 Vdc			I_{CEX}	---	20	uAdc
	2N2911	V_{CE}=150 Vdc			I_{CEX}	---	20	uAdc
3061	Emitter-base cutoff current:	Bias Cond.D V_{EB}=10 Vdc			I_{EBO}	---	10	uAdc
3076	Forward-current transfer ratio	Pulsed; 3/ I_C=1 Adc V_{CE}=4 Vdc			h_{FE}	20	60	---

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Table II. Group B inspection - (Cont'd).

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 3		15	15				
2016	Shock	Non-operating 1500G 5 blows of 0.5 msec ea. in orientations X1, Y1, Y2, Z1 (total=20 blows)			---	---	---	---
2046	Vibration fatigue	Non-operating			---	---	---	---
2056	Vibration, variable frequency	---			---	---	---	---
2006	Constant acceleration	20,000G Orientations X1, Y1, Y2, Z1			---	---	---	---
	End-Point tests:							
	Same as for subgroup 2, above							
	Subgroup 4		20	20				
2036	Terminal strength (lead fatigue)	Test Cond.E			---	---	---	---
	Subgroup 5		20	20				
1041	Salt atmosphere (corrosion)	---			---	---	---	---
	Subgroup 6		10	10				
---	Safe operating area (dc operation):	Test circuit and procedure per Fig.4 herein 4/ T_C=+100°C t=60 sec. 1 Cycle			---	---	---	---

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Table II. Group B inspection.-(Cont'd).

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 6-(Cont'd)							
		Test #1: I_C= 3 Adc, V_{CE}= 2.5 Vdc						
		Test #2: I_C= 167 mAdc, V_{CE}= 30 Vdc						
		Test #3:						
		2N2034 I_C= 125 mAdc, V_{CE}= 60 Vdc						
		2N2858 I_C= 93 mAdc, V_{CE}= 80 Vdc						
		2N2859 I_C= 75 mAdc, V_{CE}= 100 Vdc						
		2N2911 I_C= 60 mAdc, V_{CE}= 125 Vdc						
	End-Point Tests: Same as for Subgroup 2 above							
	Subgroup 7		10	10				
5/	Unclamped inductive sweep	T_A= +25°C, I_B= 0.1 Adc, I_C= 1.0 Adc				---	---	
	End-Point tests: Same as for Subgroup 2 above							
	Subgroup 8		10	10				
5/	Clamped inductive sweep	T_A=+25°C, I_C= 3.0 Adc				---	---	
	End-Point tests: Same as for Subgroup 2 above							

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Table II. Group B inspection.-(Cont'd)

Test Method Per MIL-STD-750	Examination or test 1/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 9		7	λ=5				
1031	High-temperature life (non-operating	T_{stg}=+200°C; t=340 hrs 6/			---	---	---	---
	End-Point tests:							
3041	Collector-emitter cutoff current:	Bias Cond.A V_{BE}=-1.5 Vdc						
	2N2034	V_{CE}=80 Vdc			ΔI_{CEX}	---	7/	8/
	2N2858	V_{CE}=100 Vdc			ΔI_{CEX}	---	7/	8/
	2N2859	V_{CE}=120 Vdc			ΔI_{CEX}	---	7/	8/
	2N2911	V_{CE}=150 Vdc			ΔI_{CEX}	---	7/	8/
3061	Emitter-base cutoff current:	Bias Cond.D V_{EB}=10 Vdc			I_{EBO}	---	7/	8/
3076	Forward-current transfer ratio	Pulsed; 3/ I_C=1 Adc V_{CE}=4 Vdc			Δh_{FE}	---	+30 -15%	8/
	Subgroup 10		7	λ=7				
1026	Steady state operation life:	T_A=+25°C P_T=1 W (without heat sink) t=340 hrs 6/			---	---	---	---
	2N2034	V_{CB}=40 Vdc						
	2N2858	V_{CB}=60 Vdc						
	2N2859	V_{CB}=60 Vdc						
	2N2911	V_{CB}=100 Vdc						
	End-Point tests: Same as for Subgroup 9 above							

1/See 3.4 and 4.3.2 herein.
2/Per Method 112 in Standard MIL-STD-202.
3/See 4.3.1 herein
4/See 4.3.4 herein.
5/See Fig. 6 herein.
6/See 4.2.4 herein.
7/+100% or 2 uA, whichever is greater.
8/Change from initial Group A reading.

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Table III. Group C inspection. 1/

Test Method Per MIL-STD-750	Examination or test 2/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		15	15				
1001	Barometric pressure reduced (altitude operation):	Pressure 8mmHg Normal mounting t = 60 sec, minimum			---	---	---	---
	Measurement during test:							
3041	Collector-emitter cutoff current:	Bias Cond.A V_{BE}=-1.5 Vdc						
	2N2034	V_{CE}=80 Vdc			I_{CEX}	---	20	uAdc
	2N2858	V_{CE}=100 Vdc			I_{CEX}	---	20	uAdc
	2N2859	V_{CE}=120 Vdc			I_{CEX}	---	20	uAdc
	2N2911	V_{CE}=150 Vdc			I_{CEX}	---	20	uAdc
3151	Thermal resistance	---			θ_{j-c}	---	13.3	°C/W
	Subgroup 2		λ=10	---				
1031	High-temperature life (non-operating)	T_{stg}=+200°C t=1000 hrs 3/			---	---	---	---
	End-Point tests:							
3041	Collector-emitter cutoff current:	Bias Cond.A V_{BE}=-1.5 Vdc						
	2N2034	V_{CE}=80 Vdc			I_{CEX}	---	4/	5/
	2N2858	V_{CE}=100 Vdc			I_{CEX}	---	4/	5/
	2N2859	V_{CE}=120 Vdc			I_{CEX}	---	4/	5/
	2N2911	V_{CE}=150 Vdc			I_{CEX}	---	4/	5/
3061	Emitter-base cutoff current	Bias Cond.D V_{EB}=10 Vdc			I_{EBO}	---	4/	5/

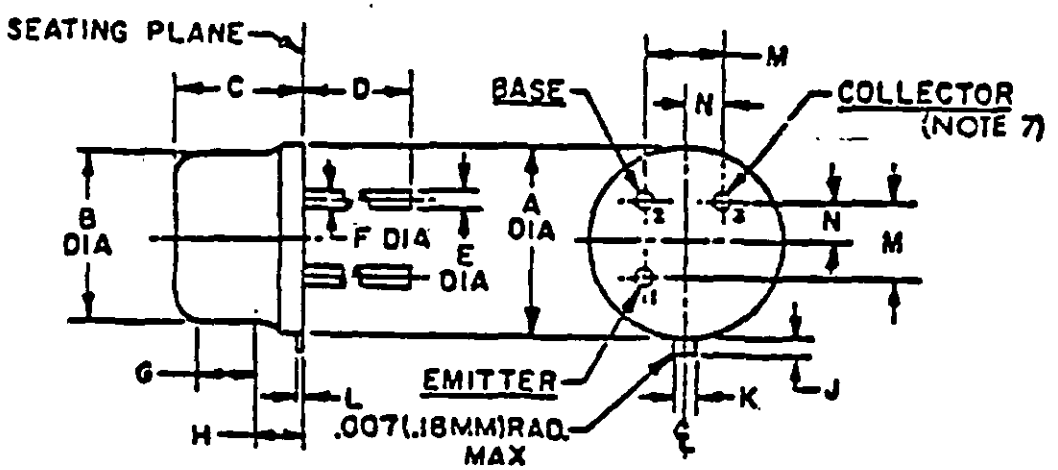
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Table III. Group C inspection. 1/ - (Cont'd).

Test Method Per MIL-STD-750	Examination or test 2/	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	End-Point tests-(Cont'd)							
3076	Forward-current transfer ratio	Pulsed; 6/ I_C=1 Adc V_{CE}=4 Vdc			Δh_{FE}	---	+30 -15%	5/
	Subgroup 3		λ=10	---				
1026	Steady state operation life:	T_A=+25°C P_T=1 W (without heat sink) t = 1000 hrs 3/			---	---	---	---
	2N2034	V_{CB}=40 Vdc						
	2N2858	V_{CB}=60 Vdc						
	2N2859	V_{CB}=60 Vdc						
	2N2911	V_{CB}=100 Vdc						
	End-Point tests: Same as for Subgroup 2, above							
	Subgroup 4		15	15				
1026	Steady state operation life:	T_C=+100 C P_T=7.5W V_{CB}=20 Vdc t = 250 hrs						
	End-Point tests: Same as for Subgroup 2 above							

1/ Periodicity: See 4.2.5 herein.
2/ See 3.4 and 4.3.2 herein.
3/ See 4.2.4 herein.
4/ ±100% or 2 uA, whichever is greater.
5/ Change from initial Group A reading.
6/ See 4.3.1 herein.

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NOTES:

1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch =25.4mm.
2. Measured in the zone beyond .250 (6.35mm) from the seating plane.
3. Measured in the zone .050 (1.27mm) and .250(6.35mm) from the seating plane.
4. Variations on Dim.B in this zone shall not exceed .010 (.25mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054 +.001 (1.37mm +.03mm) below the seating plane of the transistor, max dia leads shall be within .007 (.18mm) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance.
7. Collector internally connected to case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads.

LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.335	.370	8.51	9.40	
B	.305	.335	7.75	8.51	
C	.240	.260	6.10	6.60	
D	1.500	1.750	38.10	44.45	7,9
E	.016	.021	.41	.53	2,9
F	.016	.019	.41	.48	3,9
G	.100	---	2.54	---	4
H	---	---	---	---	5
J	.029	.045	.74	1.14	8
K	.028	.034	.71	.86	
L	.007	.125	.23	3.18	
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

Figure 1. Outline and dimensions.

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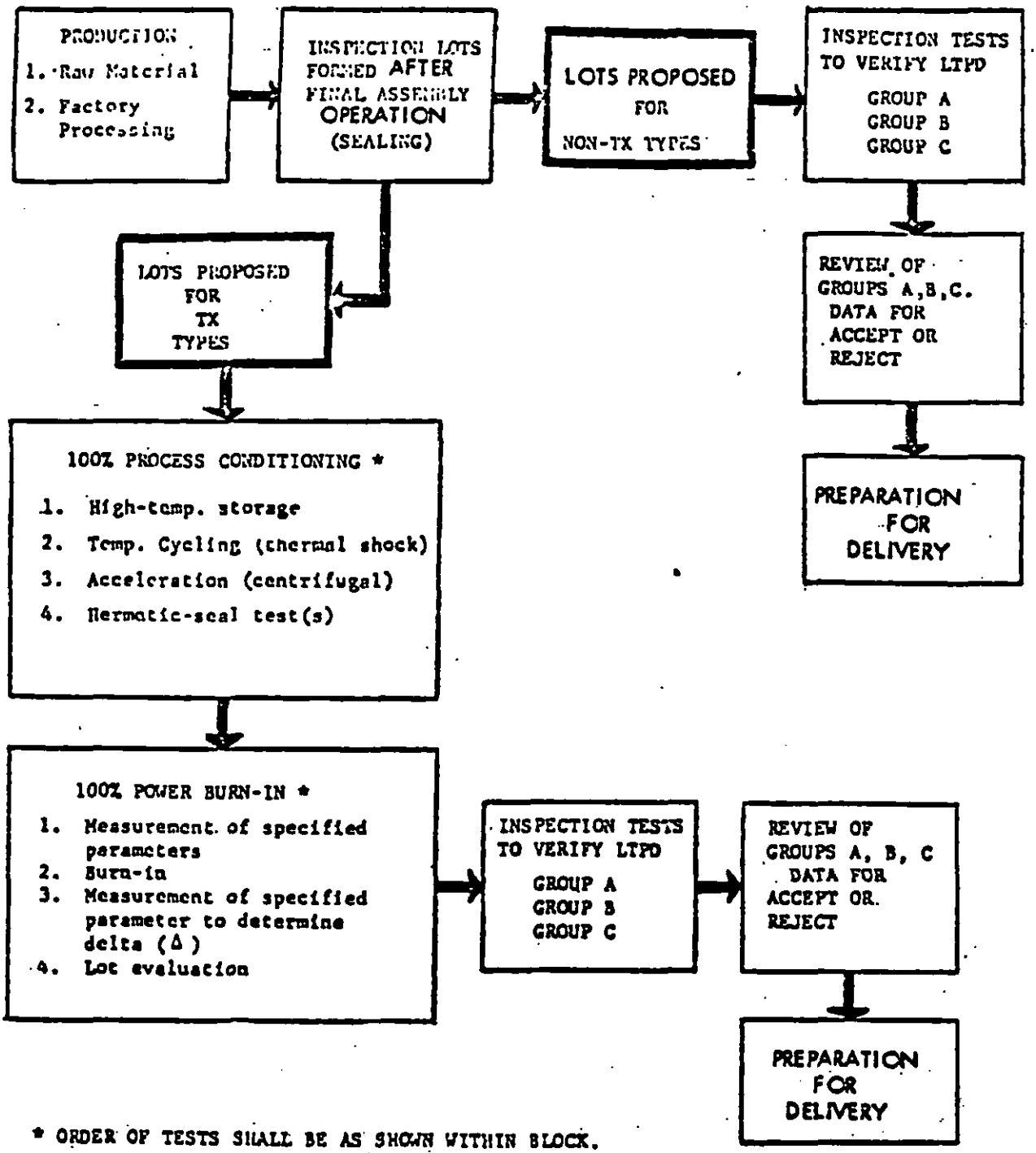
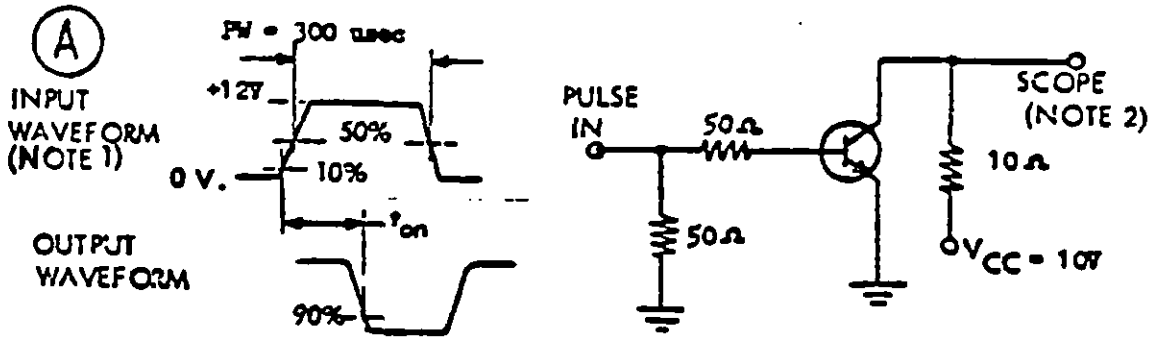


FIGURE 2. FLOW CHART OF NON-TX AND TX TYPES

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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤14 nanoseconds; duty cycle ≤ 2%; generator source impedance shall be 50Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 KΩ ; C_{in} ≤ 12 pf; rise time ≤ 5 nsec.
3. Test circuit A for Turn-On Time; test circuit B for Turn-Off Time.

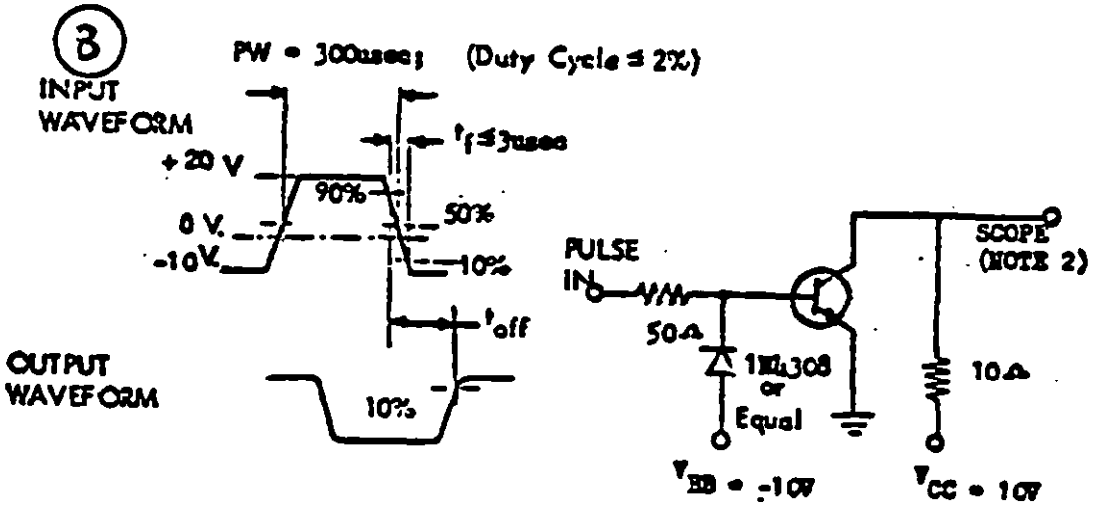
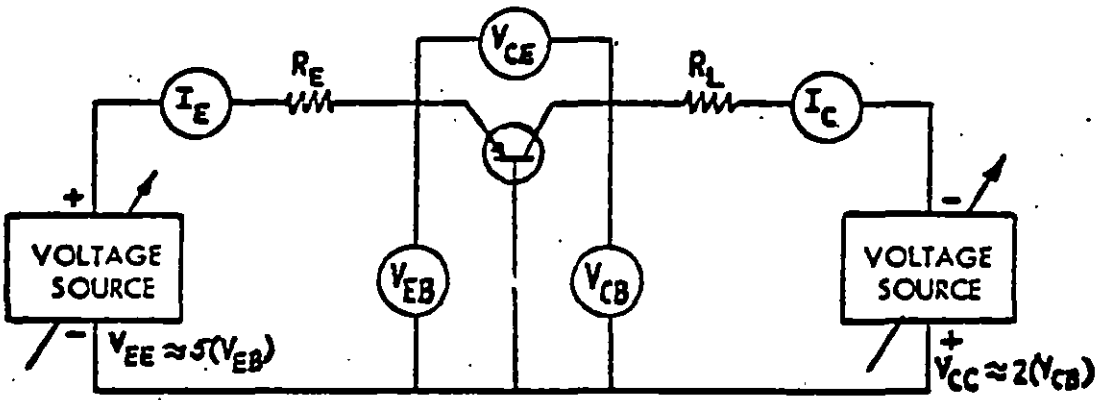


Figure 3. Pulse Response Test circuits.

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TEST PROCEDURE:

a. Starting with V_{CC} and V_{EE} at a low value, increase V_{CC} to obtain approximately the specified V_{CE}; increase V_{EE} to obtain approximately the specified I_C. Increase V_{CC} and subsequently adjust V_{EE} to obtain the specified V_{CE} and I_C. Operate the transistor at specified case temp; for the specified time duration.

b. At end of operation time: decrease V_{CC} to obtain V_{CE} near zero; turn off V_{EE}; turn off V_{CC}; remove transistor-under-test.

Fig. 4. Safe Operating Area (DC Operation) test circuit.

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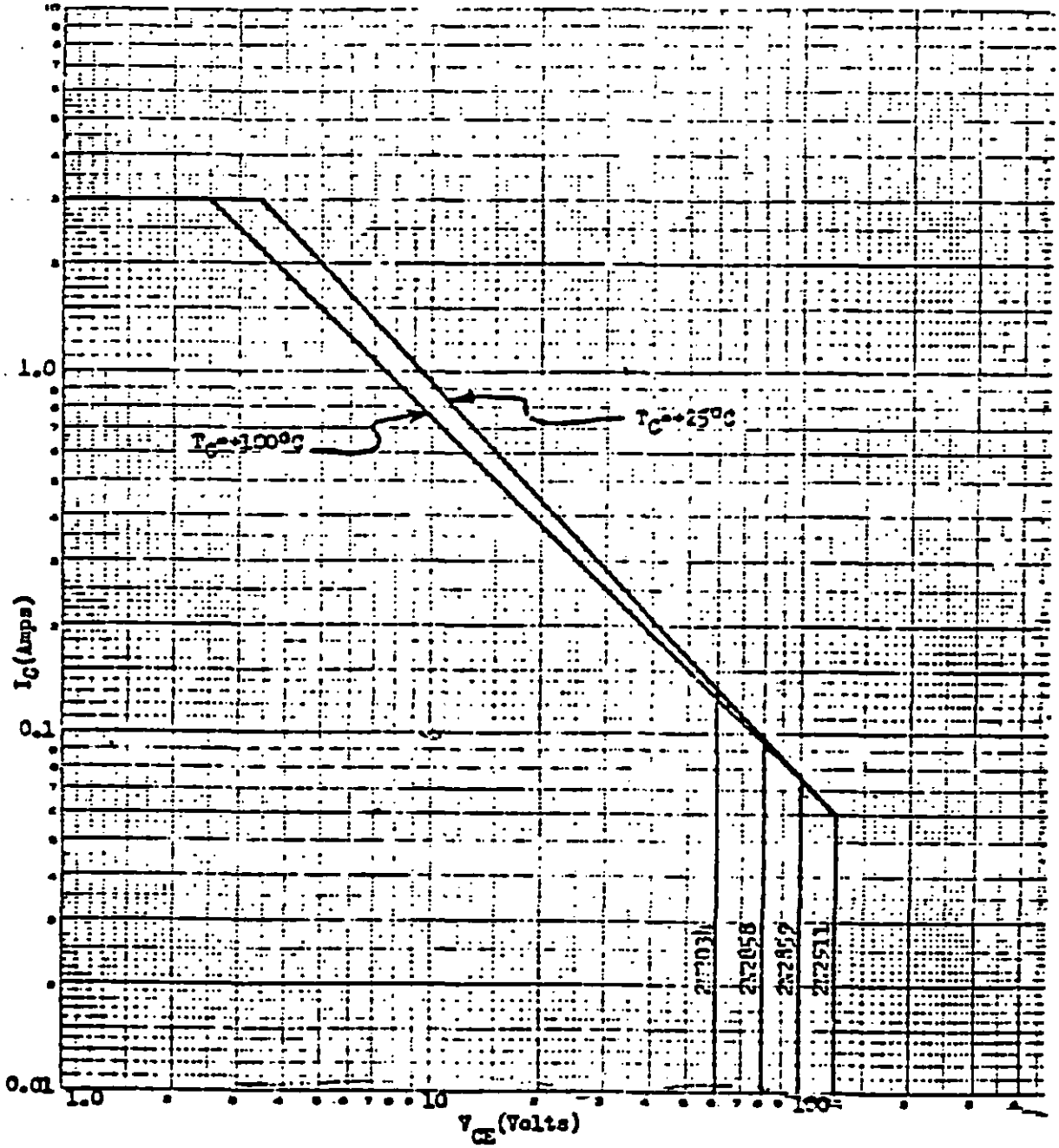
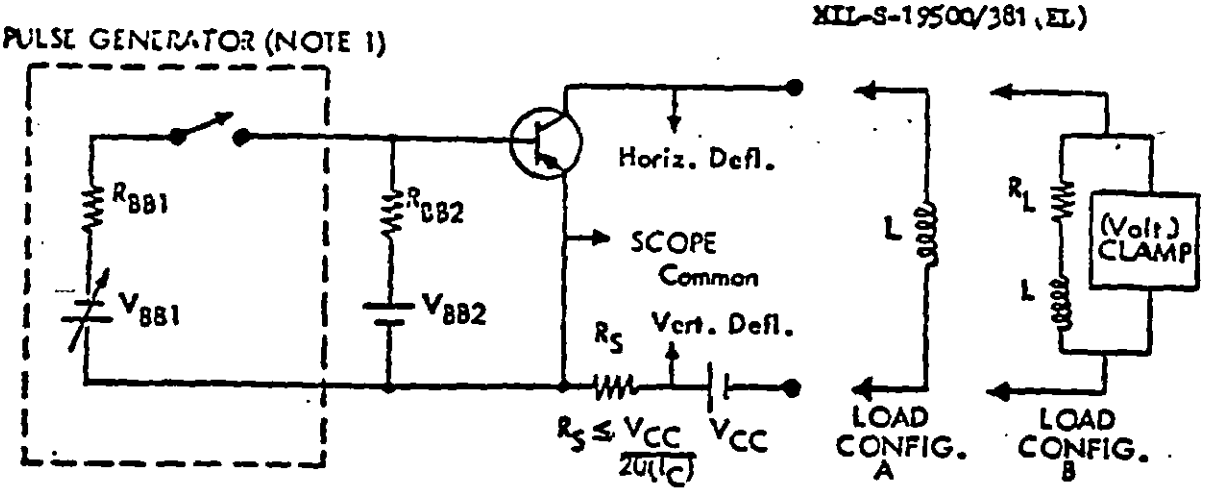


Figure 5. DC Safe Operating Area nomograph.

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NOTES:

1. Duty cycle ≤ 10%
   t = 10 msec
2. For UNCLAMPED INDUCTIVE SWEEP test:
   V_{BB1} = 5v, max
   V_{BB2} = 0
   R_{BB1} = 2Ω
   R_{BB2} = ∞
   L = 80mH; Stancor C-2691 or equal
   R_L = 0.6Ω
   R_S=0.5Ω; 12 W; 1% tol.,max; non-inductive
3. For CLAMPED INDUCTIVE SWEEP test:
   V_{BB1} = 12v, max
   V_{BB2} = 0
   R_{BB1} = 2Ω
   R_{BB2} = ∞

Clamp voltages:
2N2034-80$^{+0}_{-5}$V
2N2858-100$^{+0}_{-5}$V
2N2859-120$^{+0}_{-5}$V
2N2911-150$^{+0}_{-5}$V

   R_L = 0.15Ω
   L = 20mH; Stancor C-2691 or equal
   R_S = same as in Note 2 above

PROCEDURE:

For UNCLAMPED INDUCTIVE SWEEP test, use load circuit Configuration A.

For CLAMPED INDUCTIVE SWEEP test, use load circuit Configuration B..

Starting at a low value, adjust V_{CC} to the specified level of I_C, switch closed. Achieve specified I_B and I_C, at specified I_C, by increasing V_{BB1} or pulse-generator voltage. Observe the output waveform: I_C VS. V_{CE}. Open the switch (turn off the transistor); now, the observed trace shall be a smooth curve (between saturation and cutoff); any oscillations or trace inconsistences shall be cause for rejection; clamp voltage must be reached. Perform specified end-point tests.

Fig. 6. Unclamped and Clamped Inductive Sweep Test circuit and details.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery and the inspection of Preparation For Delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this specification.

6.2 Application guidance.- To insure proper equipment-circuit application, particular attention should be given to the differential voltage and gain characteristics, and ratings pertinent to the individual transistor types covered herein.

6.3 Ordering data.-

a. Terminal-lead length: see 3.3.3 herein.
b. Terminal-lead finish: see 3.3.4 herein.
c. Terminal-lead material: see 3.3.4.1 herein.
d. Lot inspection data: see 4.2.7 herein.

6.4 Procurement of "TX" types.- See 4.2.2 herein.

6.5 Quality assurance for "TX" types.- It is expected that where the permanent lot record, maintained as required in 4.4.1 herein, begins to show a condition of failures in excess of the LTPD rate specified herein, or a condition of entire-lot rejection (see 4.4.7.4 herein) is recorded, the Government procuring activity will determine any additional monitoring controls and evaluation criteria needed to attain restoration of the prescribed (herein) quality assurance for acceptance of product.

6.6 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, attention: AMSEL-PP-EM-2.

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